LUKAS KLEIN

Curriculum Vitae

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PROFILE

Born: 11.01.1996 – Essen
Address: Bergheimer Str. 31 / 69115 Heidelberg / Germany
Citizenship: German

EDUCATION

2022 – Today **Ph.D. in Informatics**
ETH Zürich, Information Science and Engineering Group. Supervised by Prof. Buhmann and Prof. Krause. Thesis subject: *"Knowledge Discovery through interpretable ML"*.

2022 – Today **DKFZ International PhD Program**
Helmholtz International Graduate School for Cancer Research. Includes a 3-year scholarship. Acceptance rate of 2.7% in 2022.

2022 **OxML Summer School for Machine Learning in Healthcare**
Oxford University. In person participation.

2020 – 2021 **M.Sc. in Statistics** (Master Thesis and Exchange)
ETH Zürich. Thesis subject: *"Interpretability of Disentangled Representations by Explanatory Methods"*. Final grade: Very Good.

2019 – 2021 **M.Sc. in Statistics**
University of Geneva. Final grade: Very Good.

2015 – 2018 **B.Sc. in Economics**
University of Mannheim. With minor subject Business Studies. Final grade: Good.

2018 **M.Sc. in Economics and Business Administration** (Exchange)
Norwegian School of Economics. With minor subject Business Analytics. Final Grade: Very Good.

WORK

2021 – Today **Ph.D. Candidate**
German Cancer Research Center (DKFZ) & Helmholtz Imaging, Interactive Machine Learning Group. Supervised by Dr. Jäger and Prof. Maier-Hein.

2021 **Research Intern**
ETH Zürich, Institute for Machine Learning under Prof. Buhmann.

2019 **Associate Consultant**
Camelot-ITLab GmbH.

2018 – 2019 **Thesis & Working Student for Data Science Consulting**
Camelot-ITLab GmbH. Subject: *"Pricing and Price Prediction of Glacial Acrylic Acid"*. Final Grade: Very Good.

2017 **Intern**
Microsoft National Technology Office. Co-Author of a D21 Initiative Publication for the working committee Digital Ethics: *"Einsatz von KI in der Wirtschaft"*.

2014 – 2015 **Intern**
ACE99 Ltd. and Trainingpoint Pte. Ltd. (Singapore).

VOLUNTARY

2022 – Today **Advisory Committee of Oxford ML Summer School**
AI for Global Goals.

2022 – Today **Guest Lectures / Seminars**
TU Munich, DESY, HIDA Graduate School, HIDSS4Health Graduate School.

2022 – Today **Teaching Assistant**
Advanced Topics in Machine Learning (ETH Zürich).

2022 – 2023 **Conference Reviewer**
EuroVis 2022, ICML 2023.

2016 – 2017 **Speaker for Public Relations**
AStA (Student Council) at the University of Mannheim.

2016 – 2017 **Deputy Member of the Information Committee**
University of Mannheim.

2015 – 2016 **Member of the Student Parliament**
University of Mannheim. Representative for Refugee Projects.

PUBLICATIONS

2023 **Anonymous Title: Attribution & Attention Benchmark**
Authors: Lukas Klein, Udo Schlegel, Till Bungert, Carsten Lüth, Mennatallah El-Assady, Paul Jäger, **Venue:** Under Review at ICLR.

2023 **Discovering Process Dynamic for Scalable Perovskite Solar Cell Manufacturing with Explainable AI**
Authors: Lukas Klein, Sebastian Ziegler, Felix Laufer, Charlotte Debus, Markus Götz, Klaus Maier-Hein, Ulrich W. Paetzold, Fabian Isensee, Paul Jäger, **Journal:** Advanced Materials (IF: 30) , Download here.

2023 **Understanding Solar Cell Manufacturing with Explainable AI**
Authors: Lukas Klein, Sebastian Ziegler, Felix Laufer, Charlotte Debus, Markus Götz, Klaus Maier-Hein, Ulrich W. Paetzold, Fabian Isensee, Paul Jäger, **Venue:** NeurIPS 2023 Workshop on XAI in Action (Oral, Top 4 Paper).

2023 **Towards Image-based Personalized Treatment Effect Estimation**
Authors: Shuhan Xiao, **Lukas Klein**, Paul Jäger, Jens Petersen, Klaus Maier-Hein, **Venue:** Under Review.

2023 **Navigating the Pitfalls of Active Learning Evaluation: A Systematic Framework for Meaningful Performance Assessment**
Authors: Carsten T. Lüth, Till J. Bungert, **Lukas Klein**, Paul Jäger, **Venue:** NeurIPS 2023, Download here.

2023 **A Call to Reflect on Evaluation Practices for Failure Detection in Image Classification**
Authors: Paul F. Jaeger, Carsten T. Lüth, **Lukas Klein**, Till J. Bungert, **Venue:** ICLR 2023 (Oral), Download here.

2022 **From Correlation to Causation: Formalizing Interpretable Machine Learning as a Statistical Process**
Authors: Lukas Klein, Mennatallah El-Assady, Paul Jäger, **Venue:** IJCAI 2022 Workshop on XAI, Download here.

2022 **Improving Explainability of Disentangled Representations using Multipath-Attribution Mappings**
Authors: Lukas Klein, João B. S. Carvalho, Mennatallah El-Assady, Paolo Penna, Joachim M. Buhmann, Paul Jäger, **Venue:** MIDL 2022, Download here.

SKILLS

Languages:	**German**	Native
	English	Fluent
	Chinese (Hanyu Pinyin)	Basic
Programming:	**Python**	Expert
	R	Expert
	Julia	Basic
	SQL	Advanced
	MATLAB/Octave	Basic
	CUDA	Basic
	LaTeX	Expert
	PyTorch	Expert
	TensorFlow	Basic
	JAX	Advanced
Software & Platforms:	**Linux** (Ubuntu, CentOS, SUSE Enterprise)	Advanced
	Nvidia Jetson IoT Platform	Advanced
	Singularity/Apptainer Container	Basic
	Microsoft Office (Enterprise E5 Level)	Expert
	Microsoft Power BI	Basic
	Microsoft Visual Studio Code	Expert
	Git (with GitHub & GitLab)	Advanced
	Azure (Machine Learning Studio)	Advanced
	SAP Analytics Cloud	Advanced
	IBM Spectrum LSF / Slurm	Expert
	Zoom / Microsoft Teams	Expert
	Adobe Photoshop/ InDesign	Basic
Activities:	**Cooking**, **Hiking** & **Hight Intensity Training**	



Heidelberg, 21.11.2023